



I-Beam Design

Materials:

- Carbon fiber (Toray TC275-1E/TR50S 15K)
- Vacuum bag, peel ply, release film

Machinery:

- Curing jig, tile saw, oven, 2” square steel tubes

Layup:

	Schedule	Layers	Thickness [in]
Top Flange	[90/0 _s /90/0 _s]	12	0.073
C-Section	[±45/0/±45/0/±45]	8	0.049
Core	[0/±45/0/±45/0]	7	0.042
Bottom Flange	[90/0 _s]	6	0.037

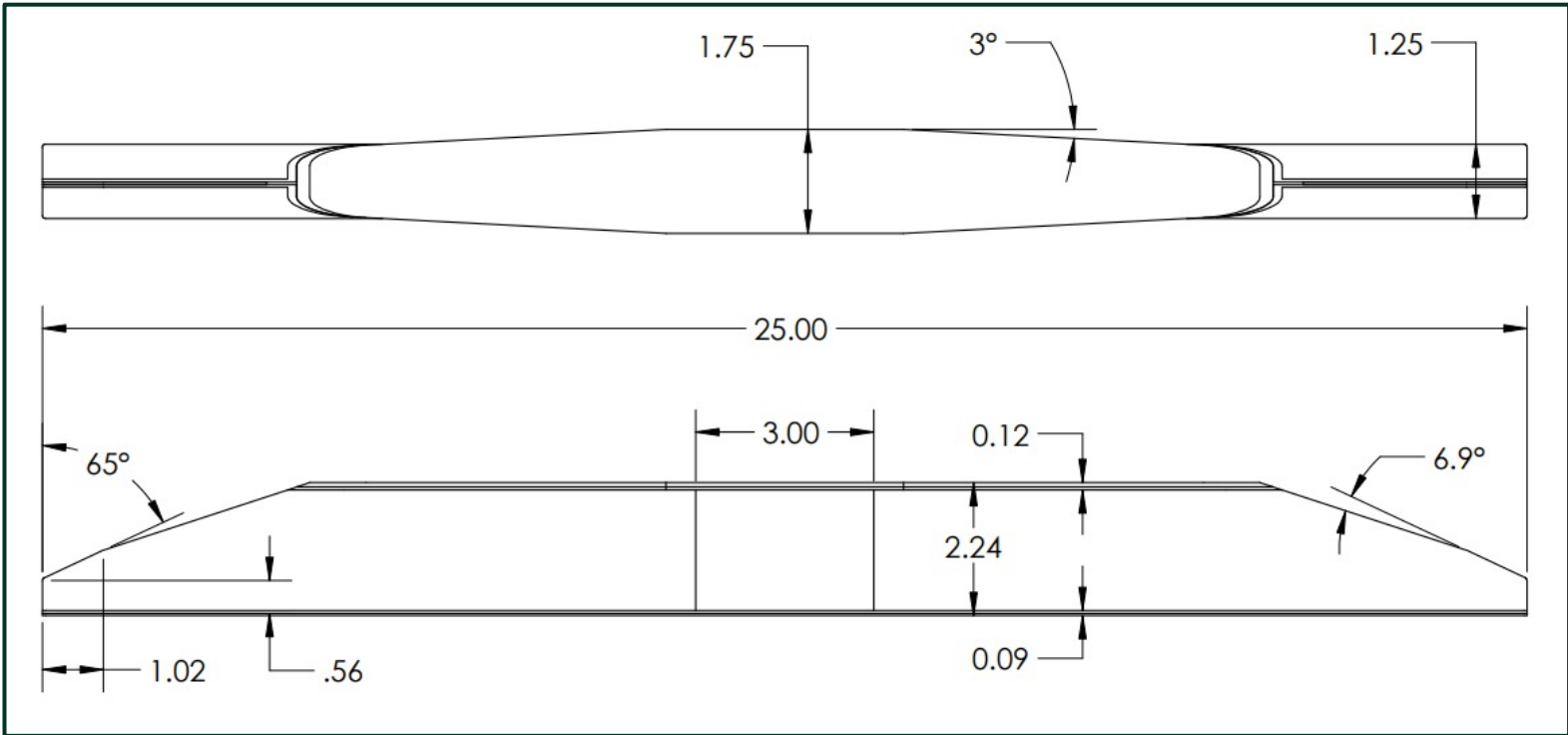


Figure 1. Dimensions of carbon fiber I-beam (front and top view).

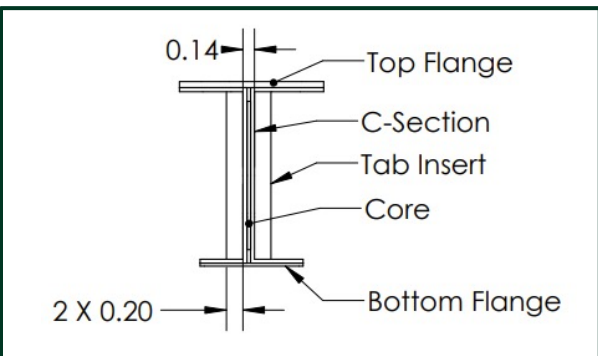


Figure 2. I-beam dimensions (side view).

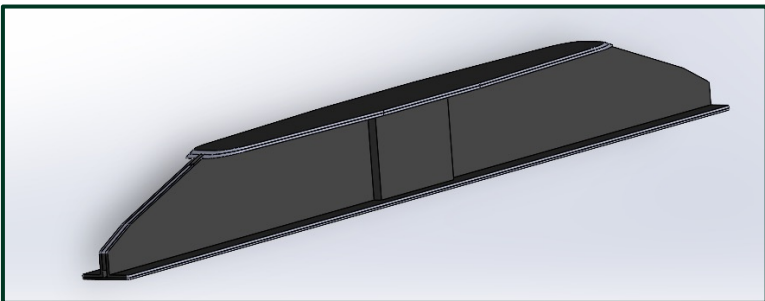


Figure 3. SolidWorks model of I-beam.

Material Characterization

Procedure:

- Tensile tested 0°, 90°, ±45° coupons in accordance with ASTM D3039/D3039M - 17.
- Determined tensile strength (σ), shear strength (τ), and shear modulus (G) from resulting data.

*L = longitudinal, T = transverse

0°		90°		±45°	
E _L [GPa]	σ_L [MPa]	E _T [GPa]	σ_T [MPa]	G _{LT} [GPa]	τ_{LT} [MPa]
105	1813	7.4	36.5	4.3	56.2

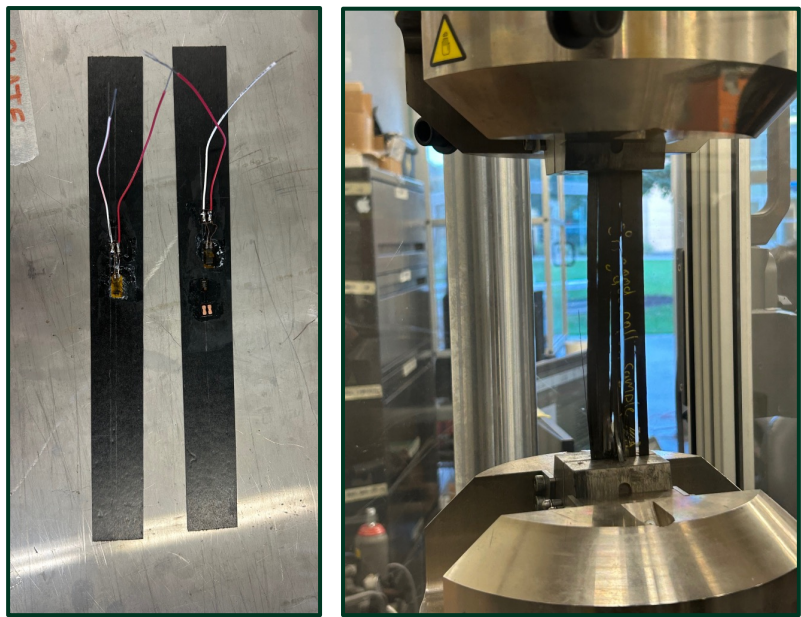


Figure 11. Coupons with strain gauges (left) and post-failure (right).

Testing

Procedure:

Three prototype beams were tested under three-point bending, resulting in a load-displacement line extending all the way to the point of failure.

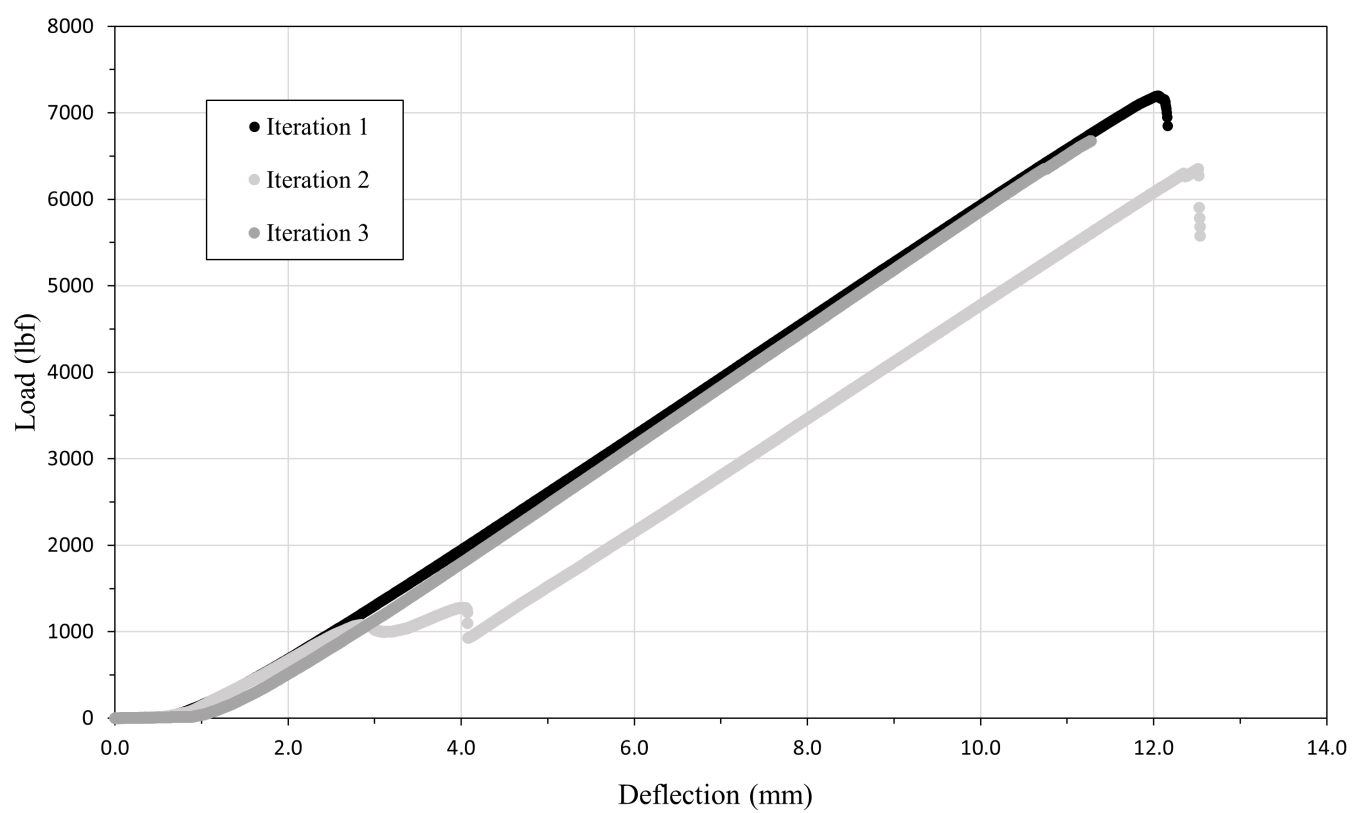


Figure 12. Three-point bend testing results for prototype beams.

Maximum Load: 7199 lbf

Maximum Deflection: 12.05 mm

Extension Rate: 1 in/min

Results:

The first beam prototype approximately met the load criteria of 7200 lbf, while weighing in 436 g. Alternate prototypes reduced weight without meeting the load criteria, fracturing along the bottom flange center line. Slight delamination was present near the flange-web interface. The final layup includes revisions necessary to improve the beam strength and reduce crack propagation.

Next Steps:

- Abaqus simulation to remove excess layers
- Improve jig clamping to ensure even pressure
- Reduce web thickness
- Iteratively test tab geometry and placement

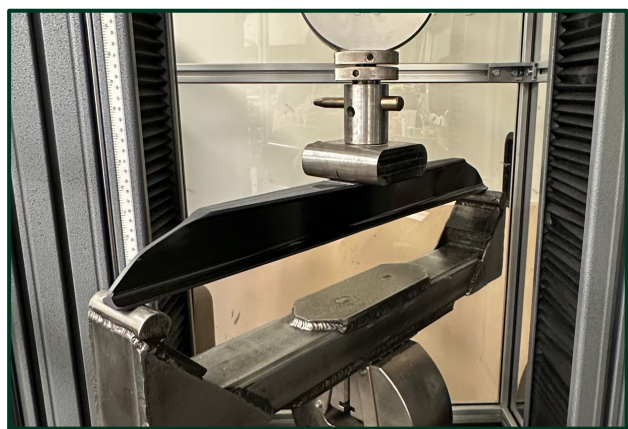


Figure 13. Three-point bend test apparatus.

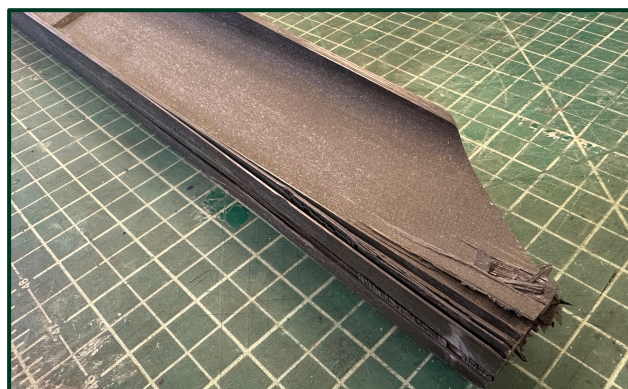


Figure 14. Beam failure and crack propagation.

Manufacturing Process



Figure 4. Cut layers of carbon fiber according to schedule.

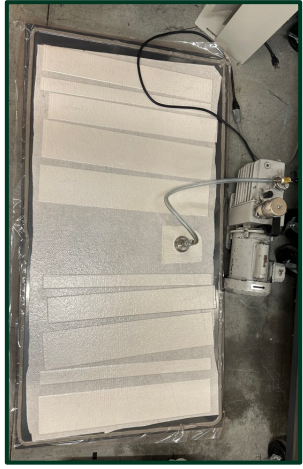


Figure 5. Debulk and stack layers.

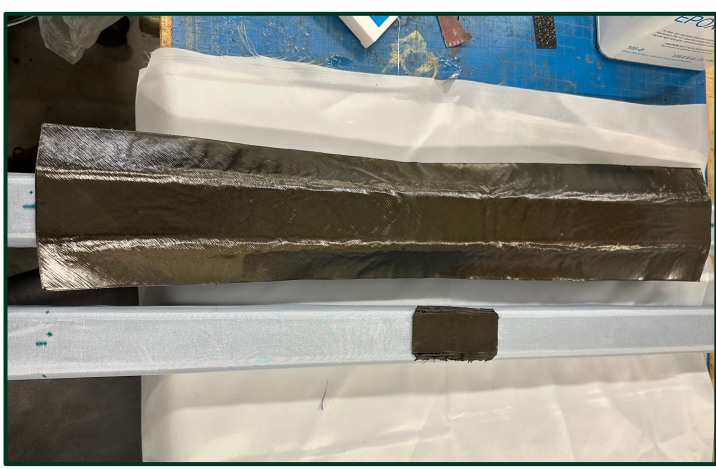


Figure 6. Apply layers and tab inserts to metal tubing.

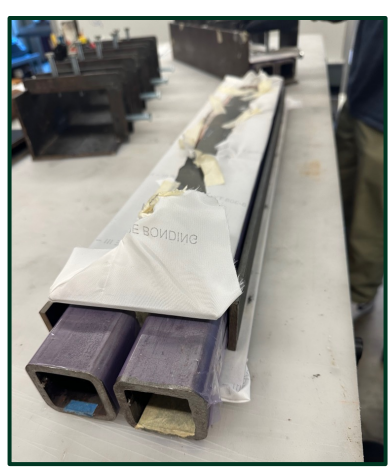


Figure 7. Compress tubes with peel-ply plates.

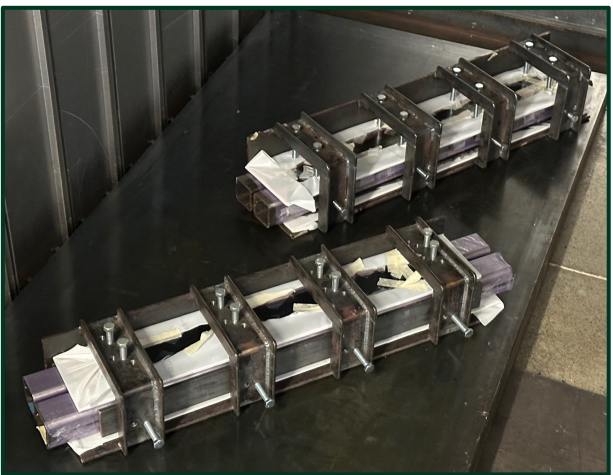


Figure 8. Cure beam + jig in oven, ramping to 265°F.



Figure 9. Extract beam from tubing and make first cuts.

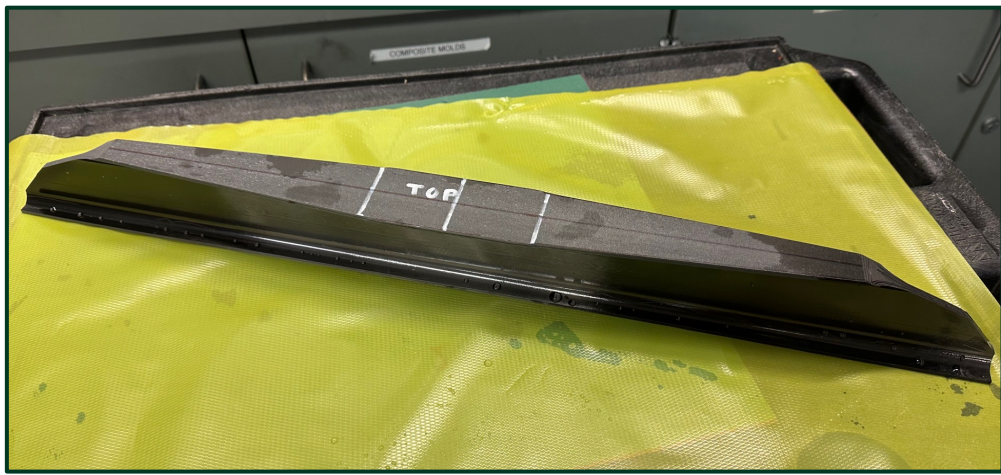


Figure 10. Cut and sand beam down to desired dimensions.

Acknowledgements

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